

ECE/CS 757 Advanced Computer Architecture II

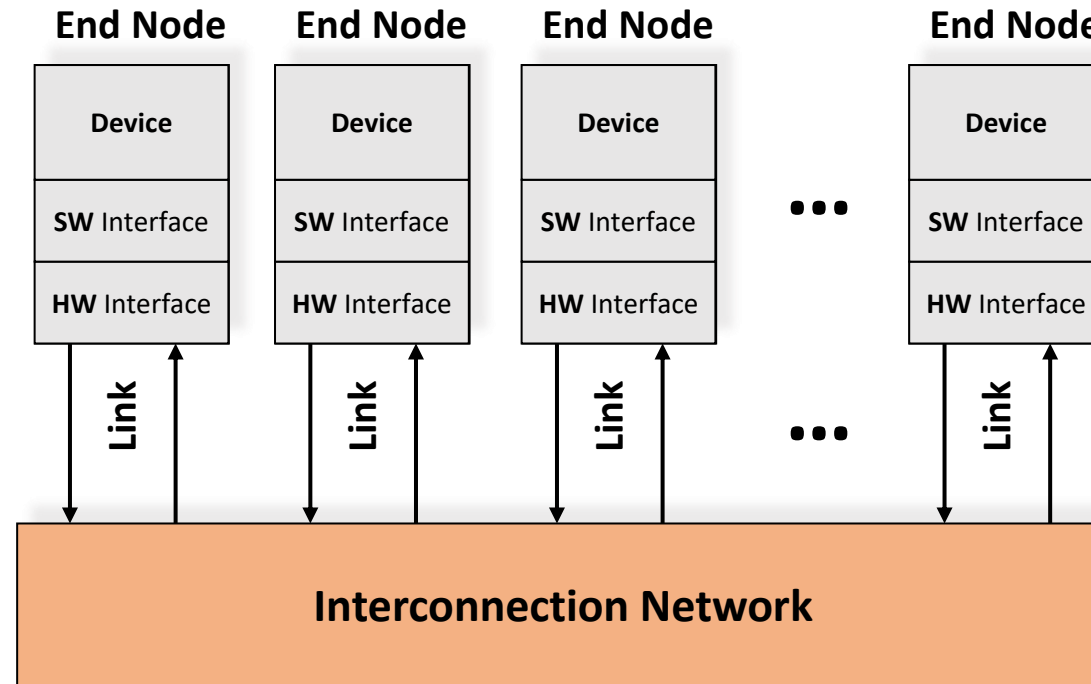
Spring 2025

Interconnects

Instructor: Prof. Joshua San Miguel

Some slides adopted from Natalie Enright Jerger

Interconnection Networks



Device (or Node):

- Component within a computer
- Single computer
- System of computers

Interconnection Networks

The cost of communication is a necessary evil that only worsens as we scale to more and more nodes.

Interconnection networks need to:

- Transfer the maximum amount of information
- In the least amount of time
- Within cost constraints (e.g., power, area)

Internetworking:

Interconnection of multiple networks

Types of Interconnection Networks

Networks-on-Chip (NoCs):

- Interconnection of functional units, caches, directories, processor cores, etc.
- 10s-100s of nodes in mm distances

System/Storage Area Networks (SANs):

- Interconnection of processors, memory, storage devices, etc.
- 100s of nodes in ~100m distances (e.g., InfiniBand)

Local Area Networks (LANs):

- Interconnection of autonomous computer systems
- 100s-1000s of nodes in ~10km distances (e.g., Ethernet)

Wide Area Networks (WANs):

- Interconnection of LANs distributed across the world
- Millions of nodes in ~100,000km distances

Types of Interconnection Networks

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this course

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Network-On-Chip – Evolution

Ad hoc wiring:

- Arbitrary connections, small number of nodes
- High complexity, area, latency

Buses:

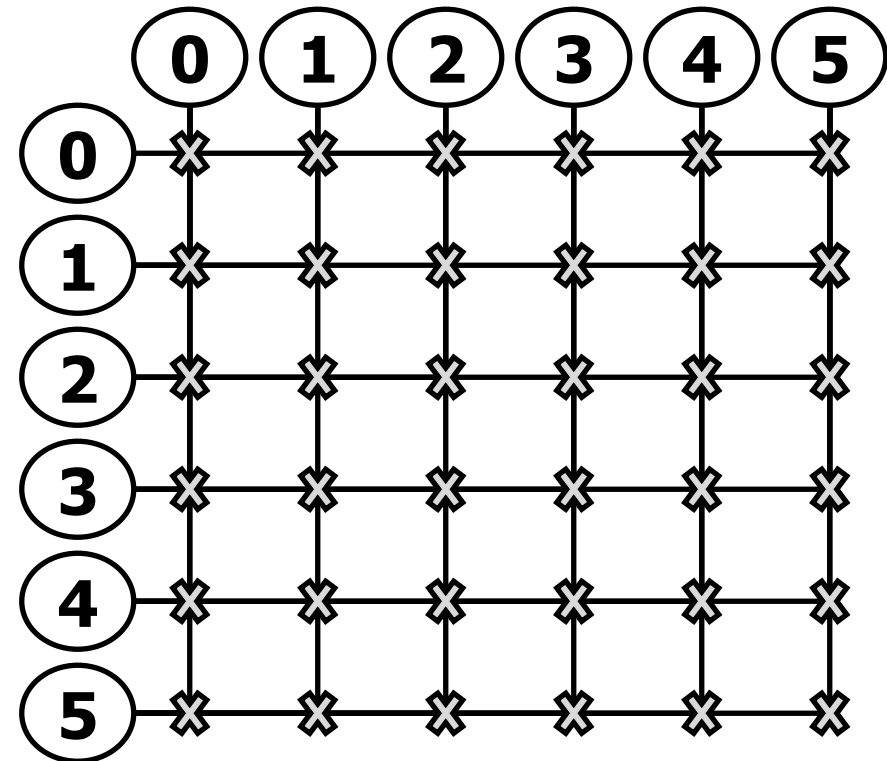
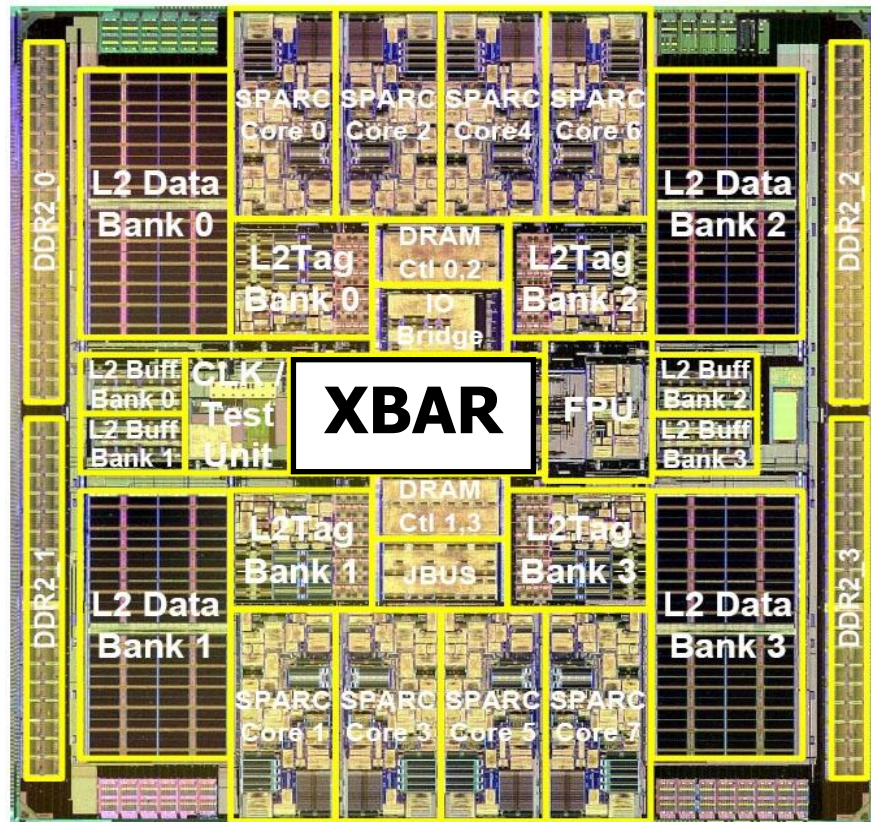
- Serialized traffic
- Low throughput with more nodes

Crossbars:

- High throughput for distinct source-destination traffic
- High area and power with more nodes

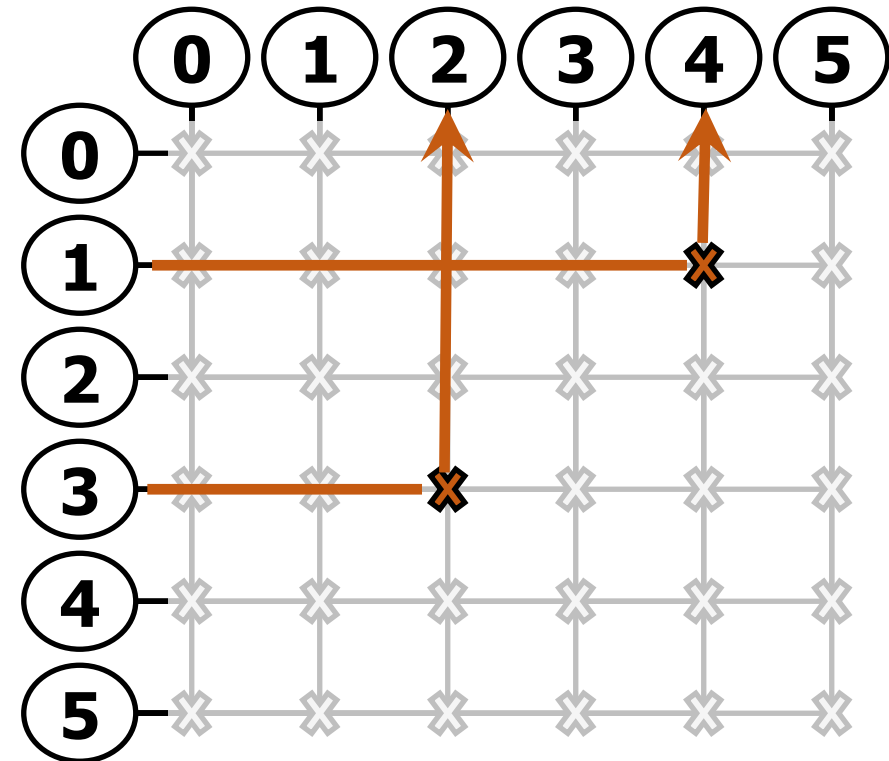
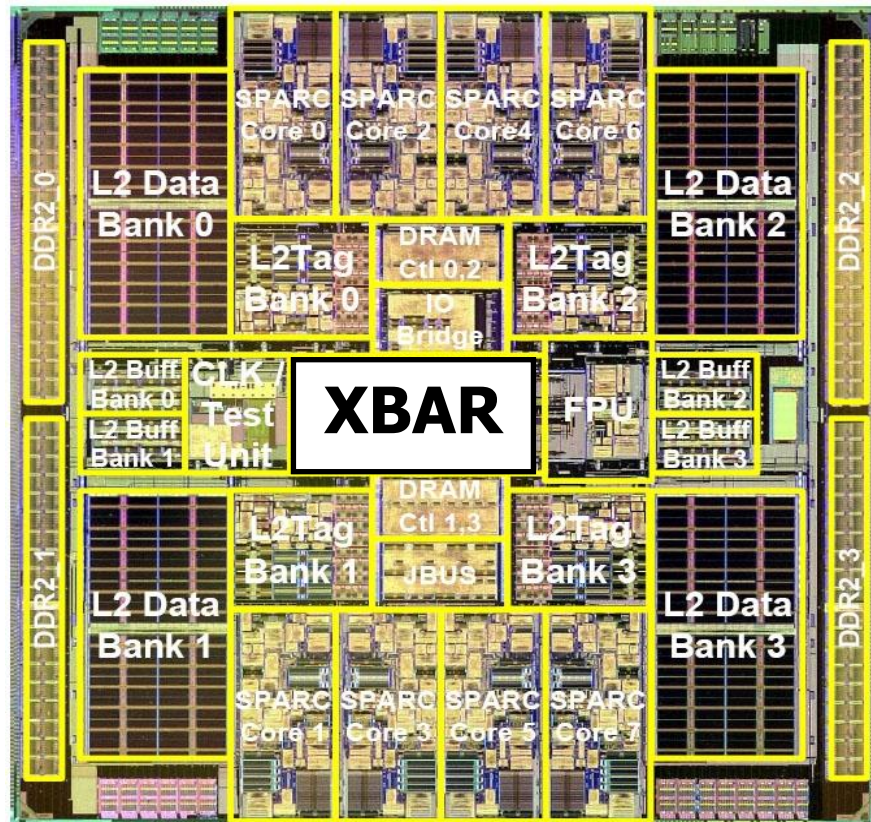
Network-On-Chip – Evolution

e.g., Sun Niagara



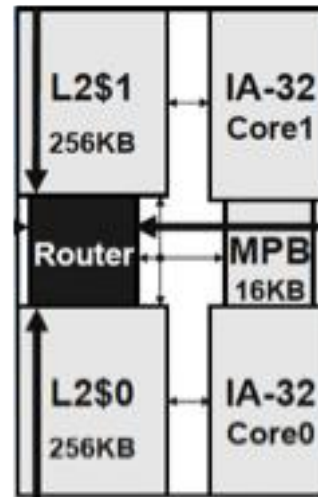
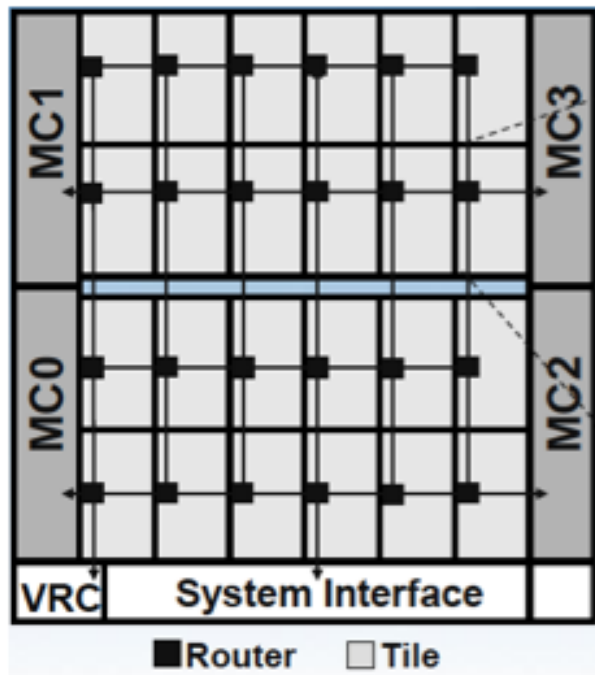
Network-On-Chip – Evolution

e.g., Sun Niagara



Network-On-Chip – Evolution

e.g., Intel Single-Chip Cloud Computer (SCC), 2D mesh

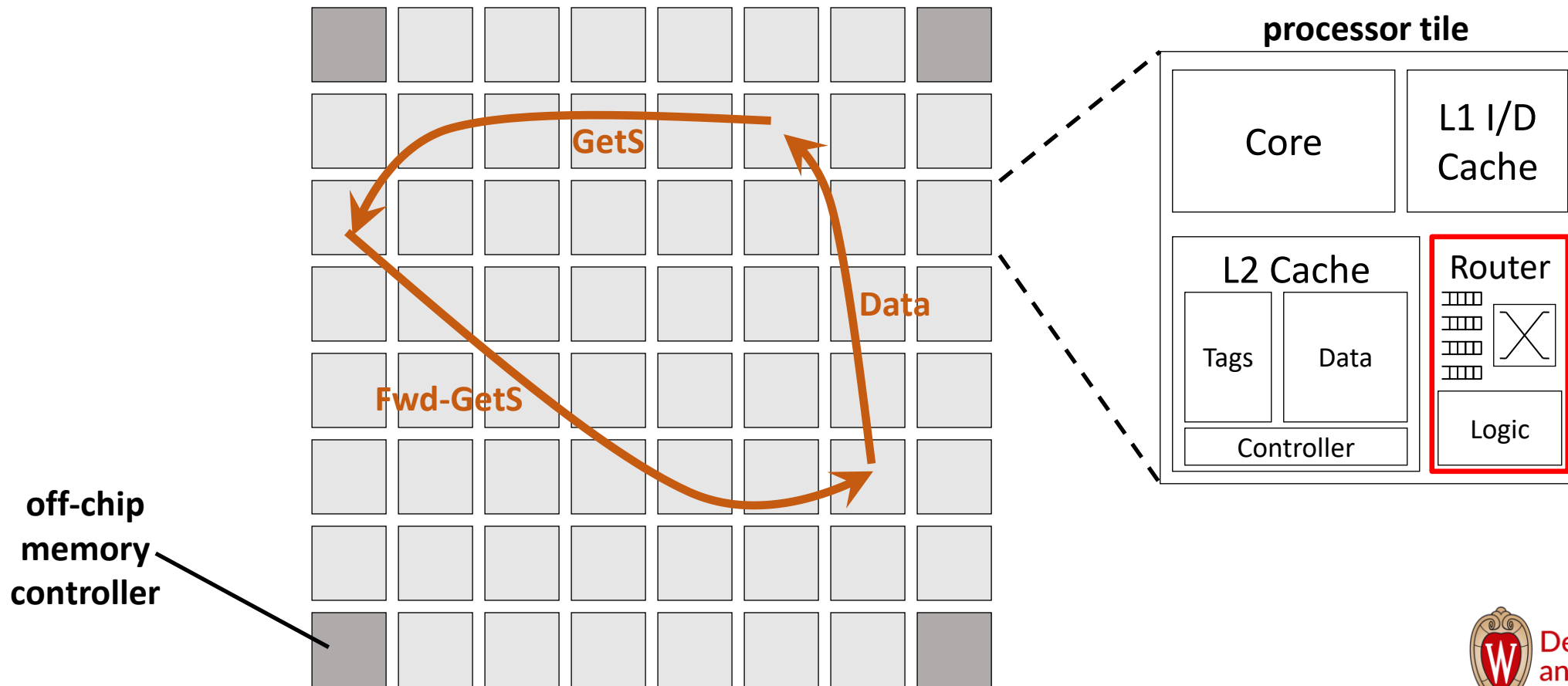


Courtesy: Jason Howard, Intel

Network-On-Chip – System Interface

Focus on NoCs for cache-coherent shared memory CMPs

- Network traffic is made up of coherence messages



Network-On-Chip – System Interface

Different message types:

- Unicast
- Multicast
- Broadcast

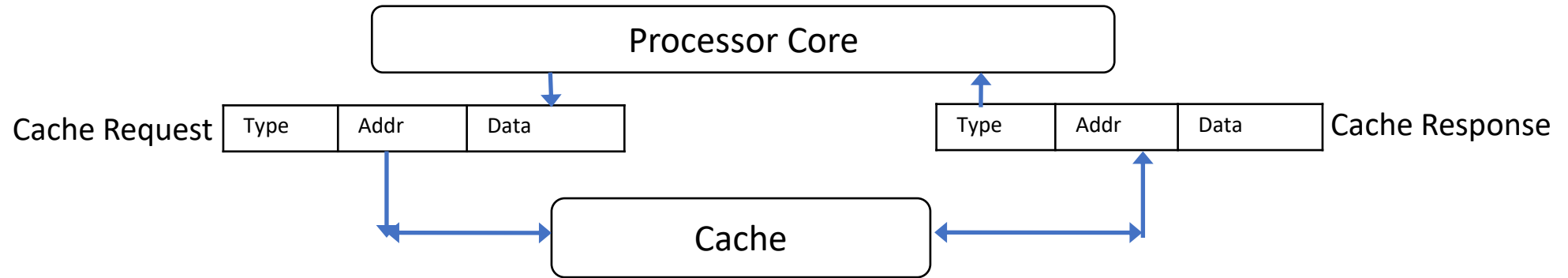
Snooping-based coherence:

- Majority of messages: broadcast
- High bandwidth demands
- Often rely on network ordering

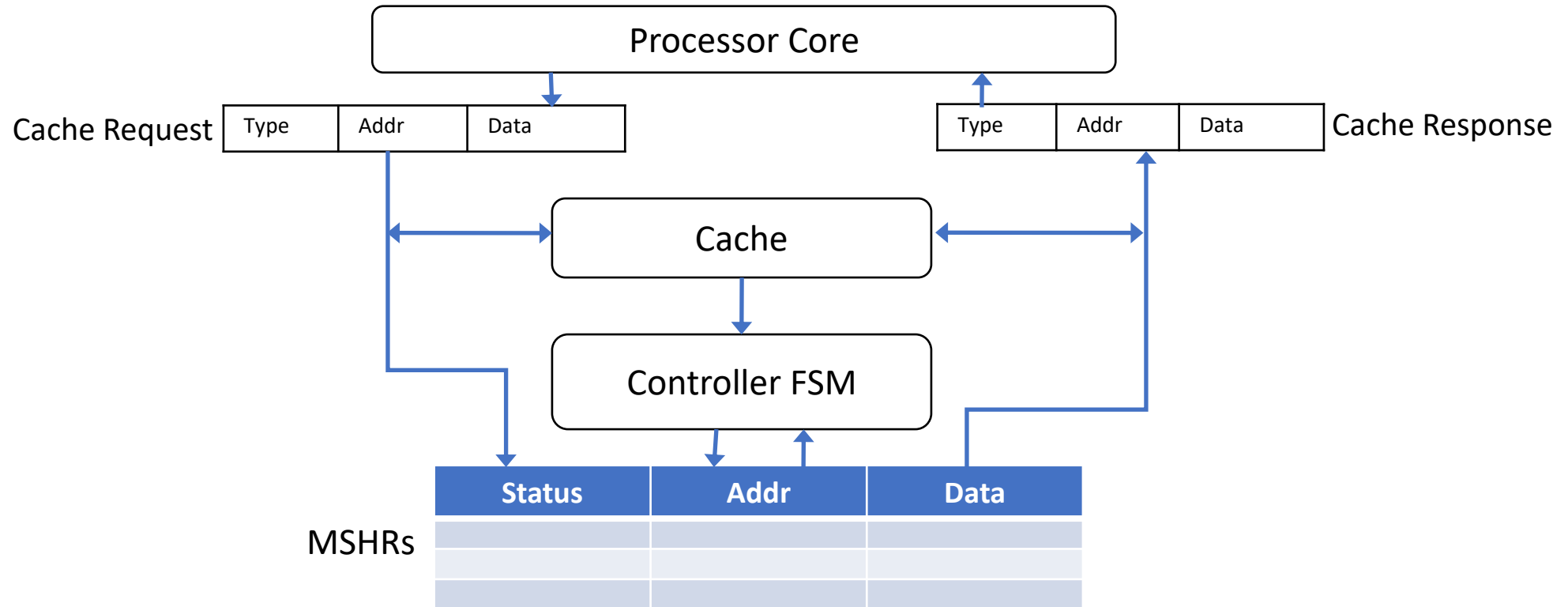
Directory-based coherence:

- Majority of messages: unicast
- Low bandwidth demands (but requires low latency)
- Point-to-point communication scales better with more cores

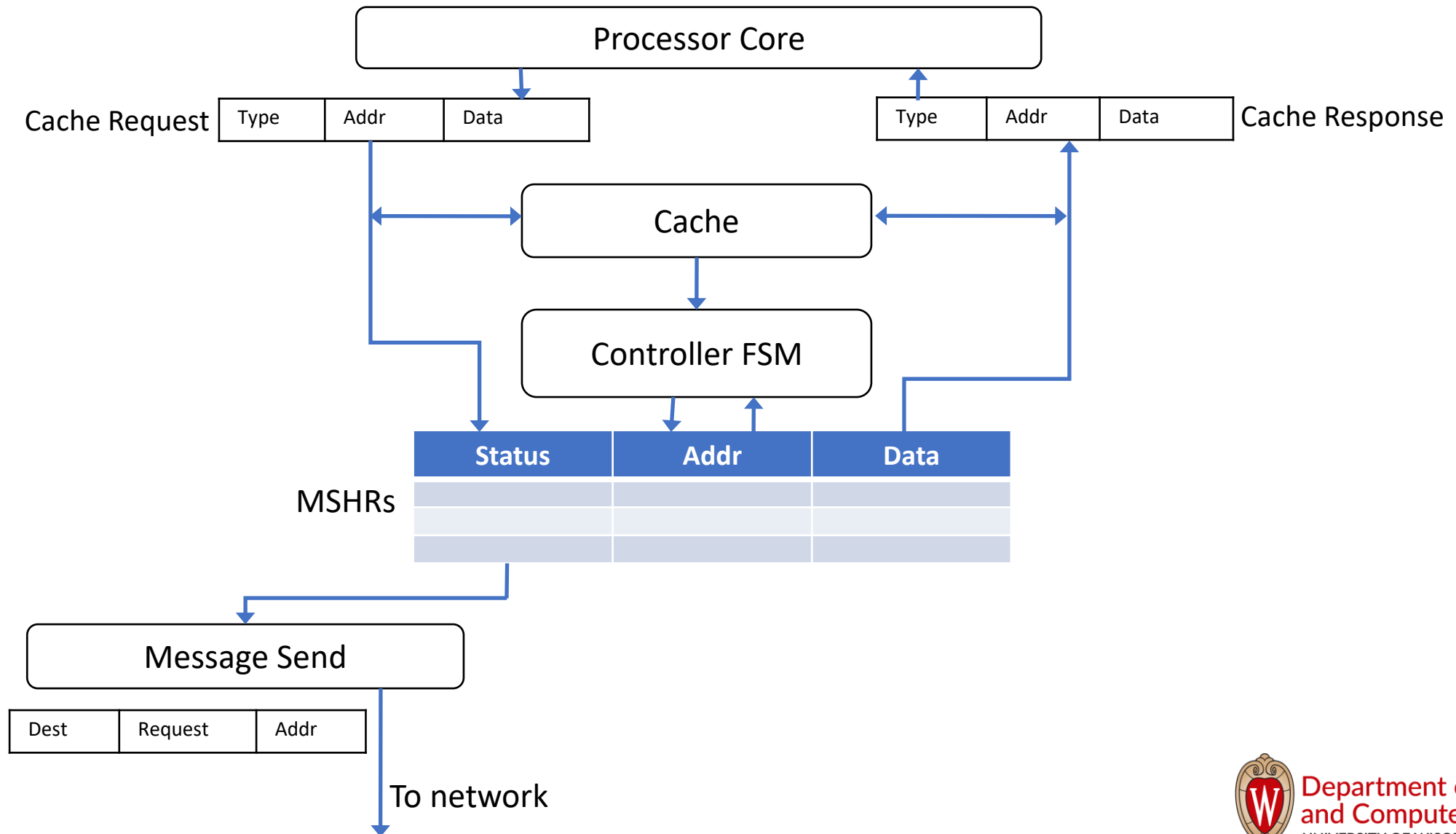
Network-On-Chip – System Interface



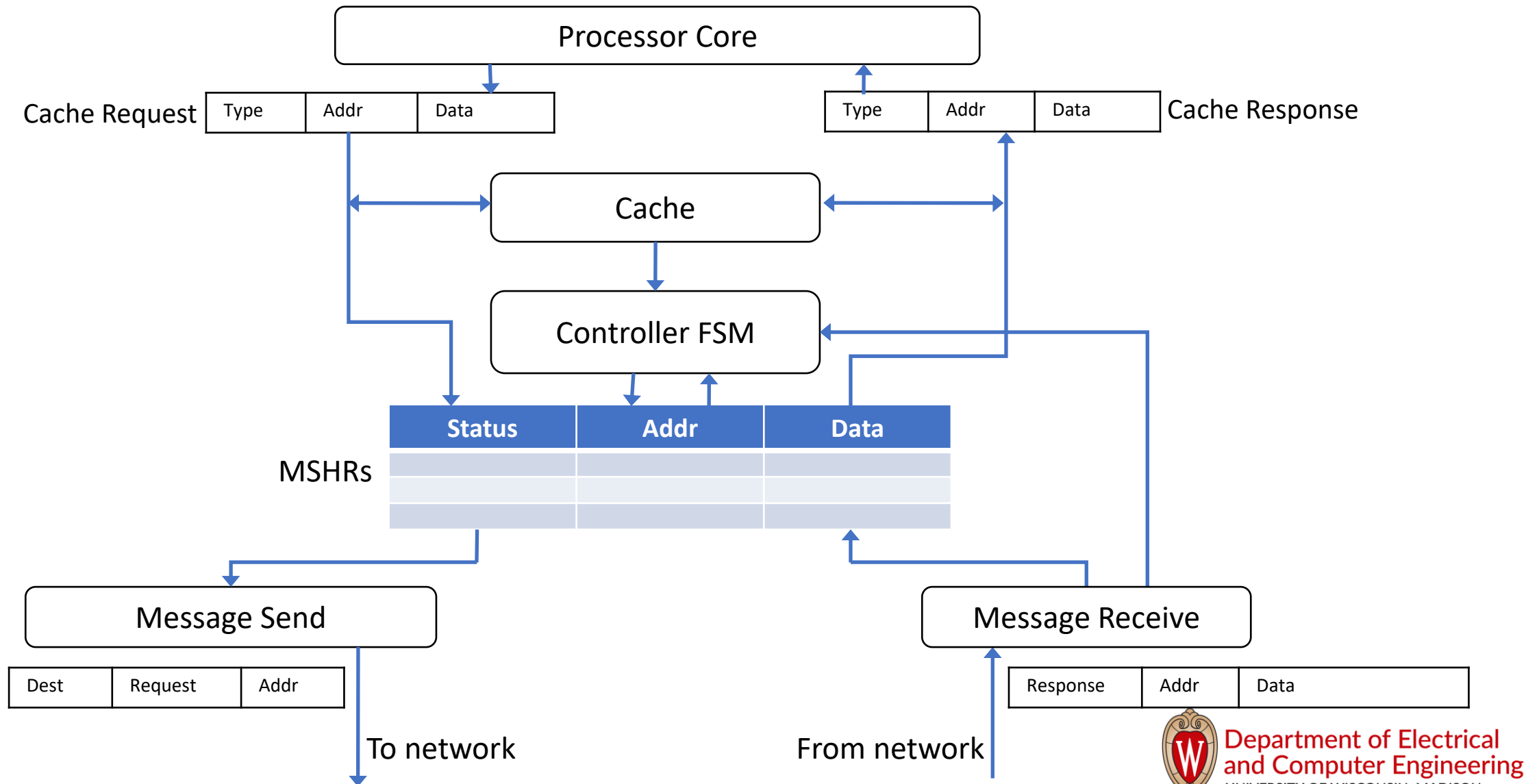
Network-On-Chip – System Interface



Network-On-Chip – System Interface



Network-On-Chip – System Interface



Network-On-Chip

Key design decisions:

Topology

- Where to place routers/nodes and how to link them together

Routing

- Which set of links to traverse between source-destination pairs

Flow Control

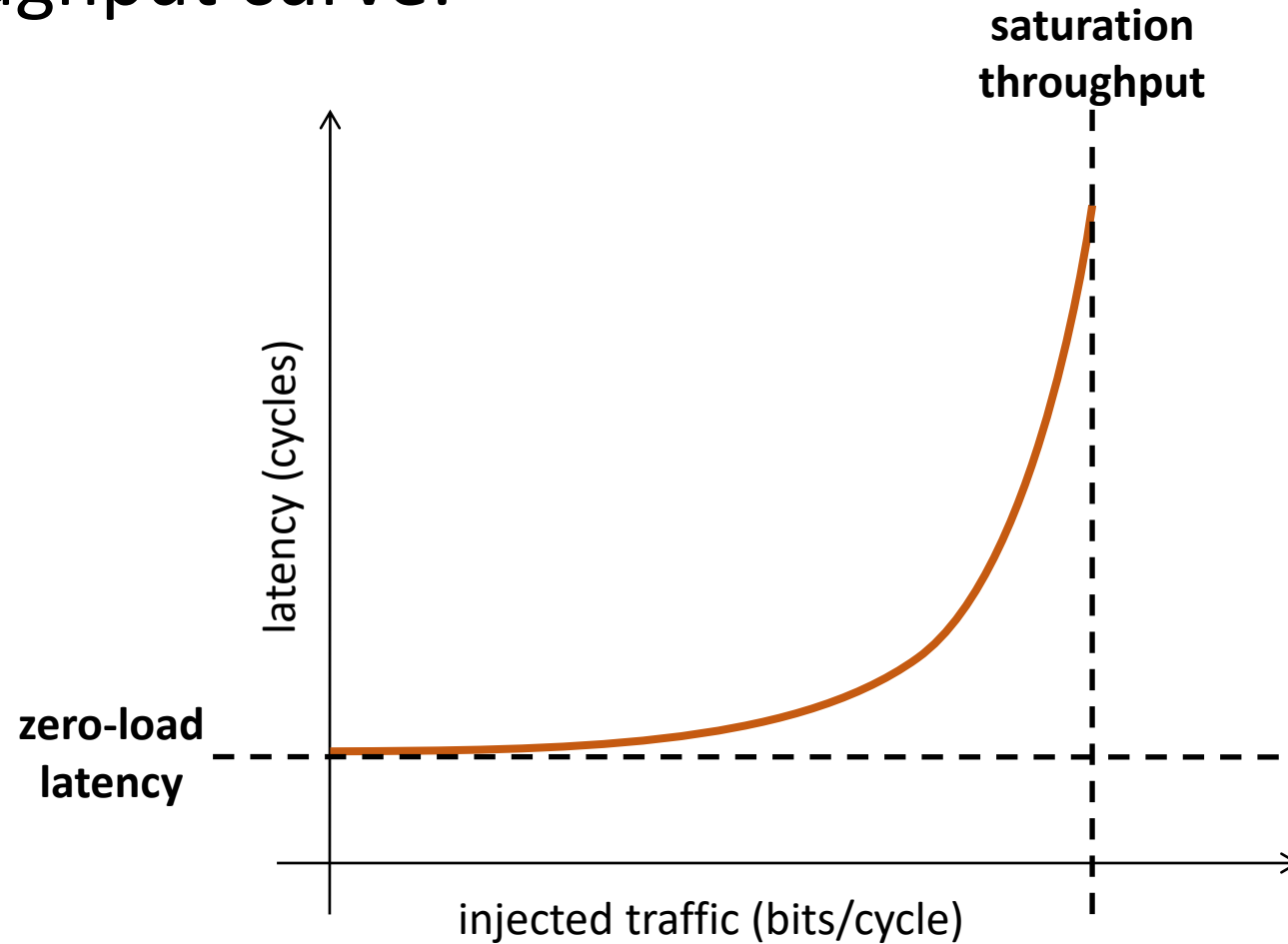
- What granularity to move messages through links and routers

Router Microarchitecture

- How to pipeline and arbitrate incoming messages in routers

Network-On-Chip – Evaluation

Latency-throughput curve:



Network-On-Chip – Evaluation

Metrics for evaluating NoC performance and cost/complexity:

- Degree
- Hop count
- Network diameter
- Maximum channel load
- Bisection bandwidth
- Path diversity

Network-On-Chip – Degree

Degree:

Number of (bidirectional) links going in/out of a router to other routers

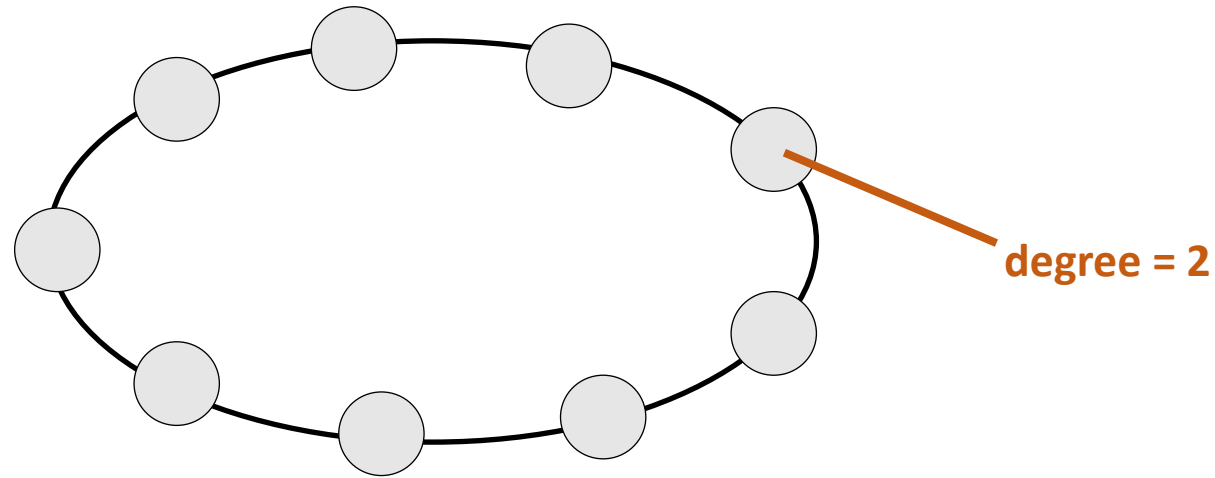
- Proxy for estimating cost (i.e., higher degree requires more ports per router)
- Affected by topology

Network-On-Chip – Degree

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Number of (bidirectional) links going in/out of a router to other routers

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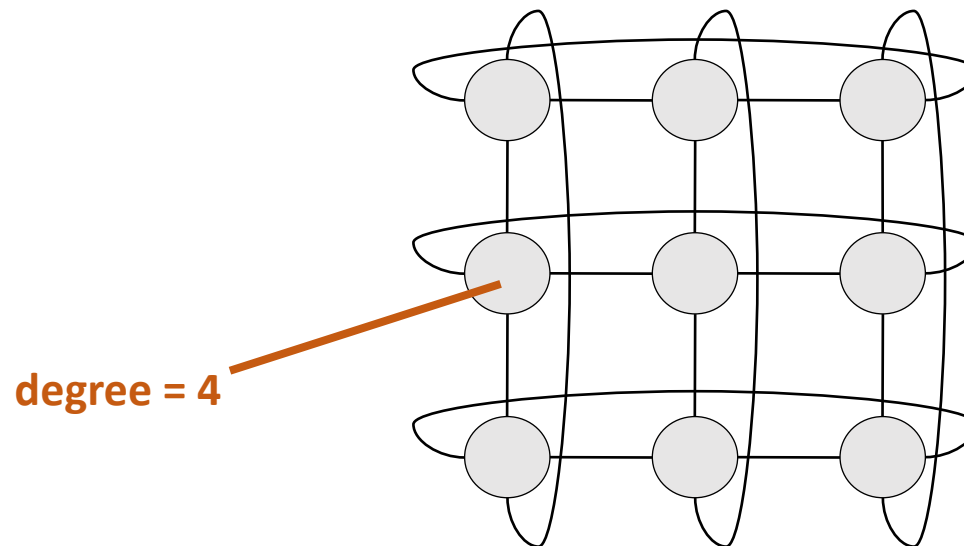


Network-On-Chip – Degree

Degree:

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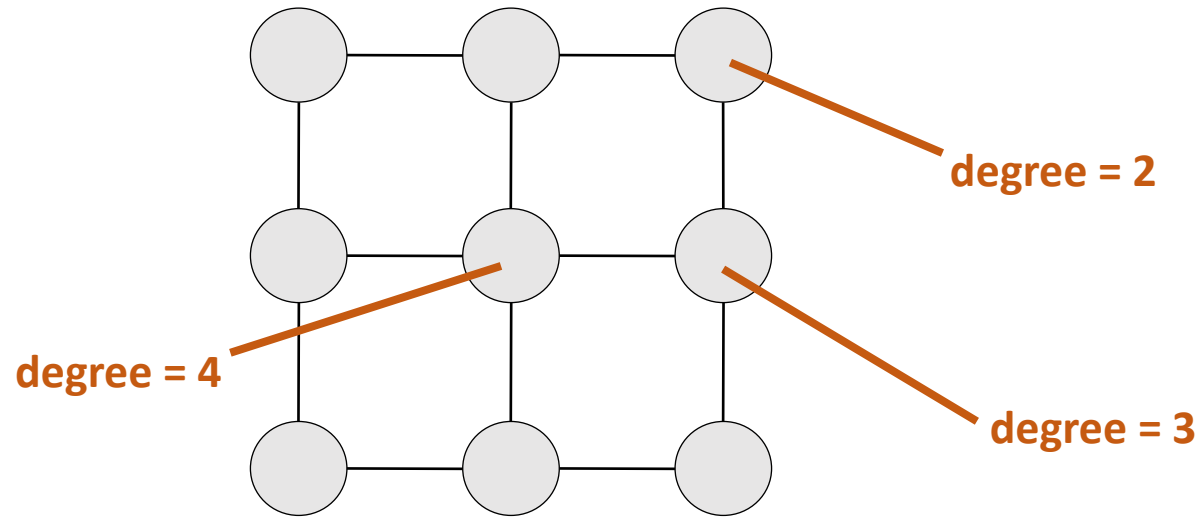


Network-On-Chip – Degree

Degree:

Number of (bidirectional) links going in/out of a router to other routers

- Proxy for estimating cost (i.e., higher degree requires more ports per router)
- Affected by topology



Network-On-Chip – Hop Count

Path:

Ordered set of links traversed by packet from source to destination

Hop Count:

Number of links traversed by packet from source to destination

- Proxy for estimating latency
- Affected by topology and routing

Minimal Hop Count:

Smallest hop count connecting two nodes

- Quickest possible route between source and destination

Network-On-Chip – Network Diameter

Average Minimal Hop Count:

Average of every source-destination pair's minimal hop count

Network Diameter:

Largest minimal hop count in the network

- Pair of source and destination nodes that are furthest apart

Network-On-Chip – Network Diameter

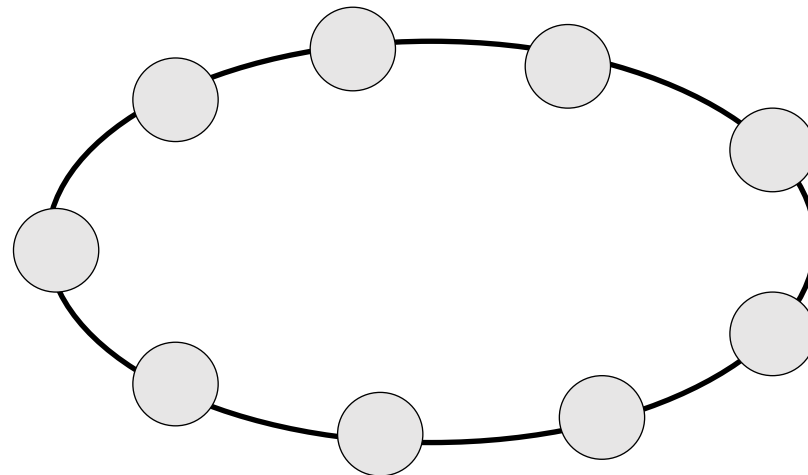
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Network diameter?

Network-On-Chip – Network Diameter

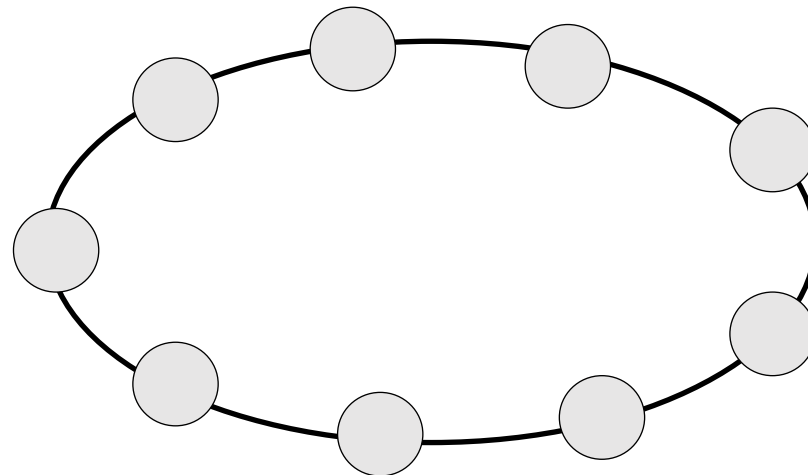
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Network diameter? 4

Network-On-Chip – Network Diameter

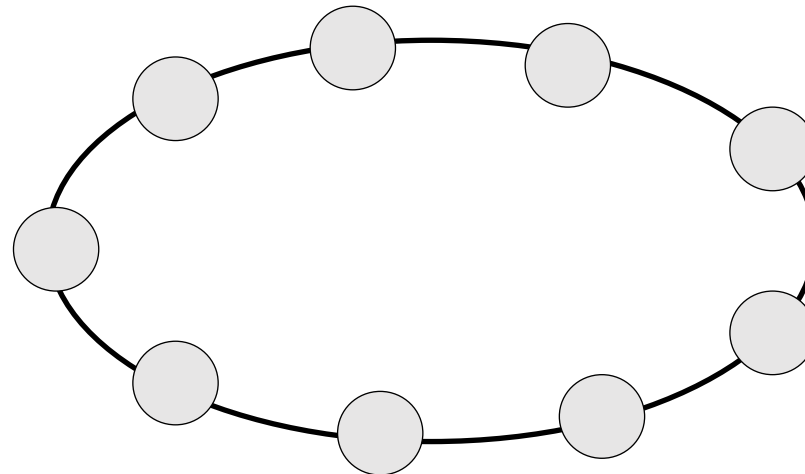
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Network diameter? **4**

Avg min hop count? **2.22**

Network-On-Chip – Network Diameter

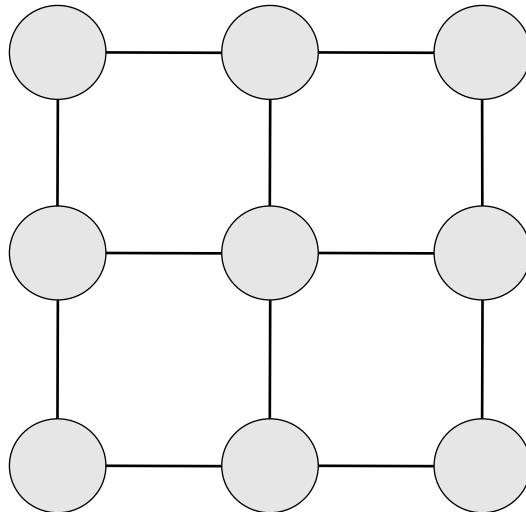
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Network diameter?

Network-On-Chip – Network Diameter

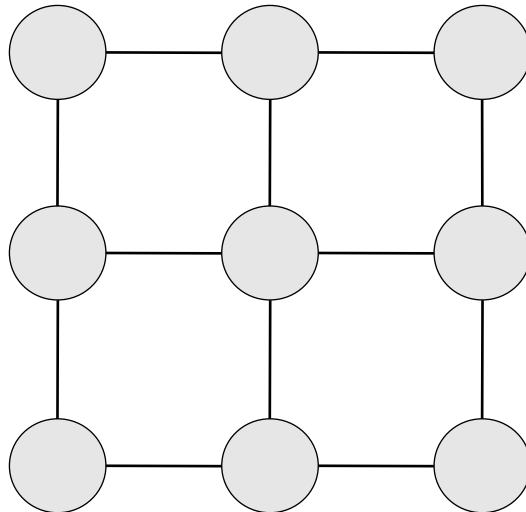
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Network diameter? 4

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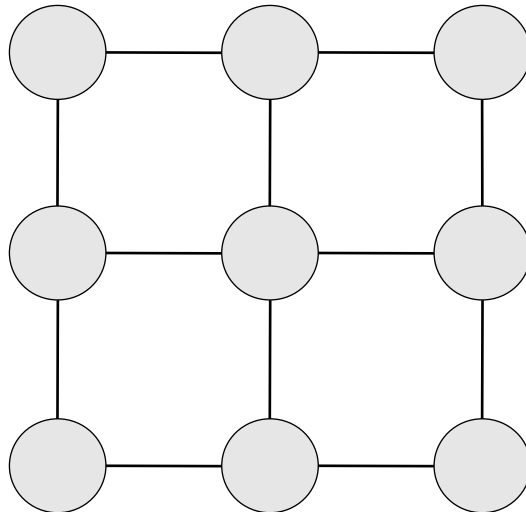
Average Minimal Hop Count:

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Network Diameter:

Largest minimal hop count in the network

- Pair of source and destination nodes that are furthest apart



Network diameter? 4

Avg min hop count? 1.77



Network-On-Chip – Network Diameter

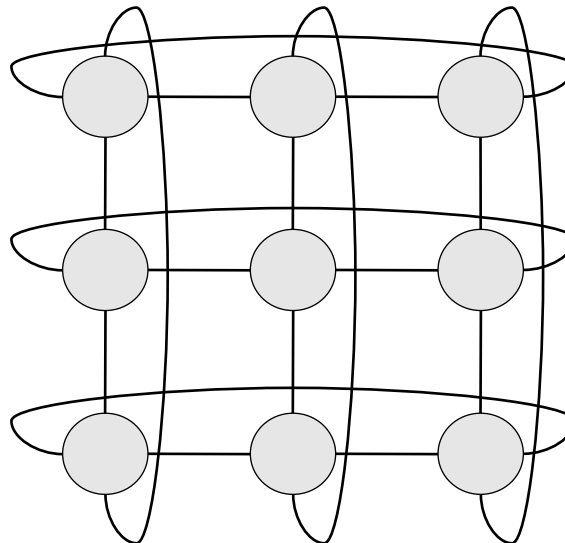
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Network diameter?



Network-On-Chip – Network Diameter

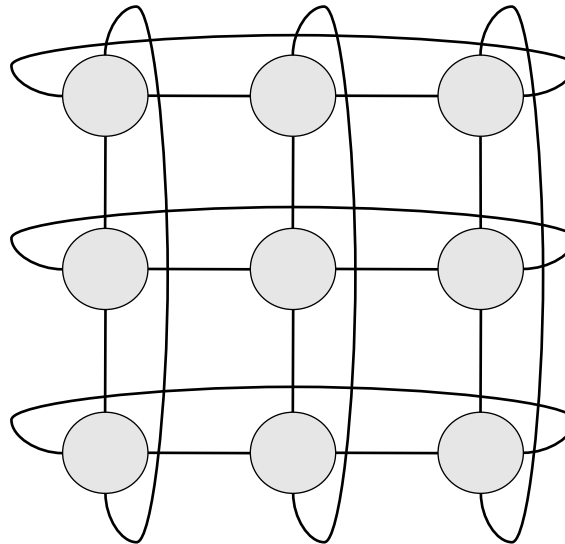
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Network diameter? 2

Network-On-Chip – Network Diameter

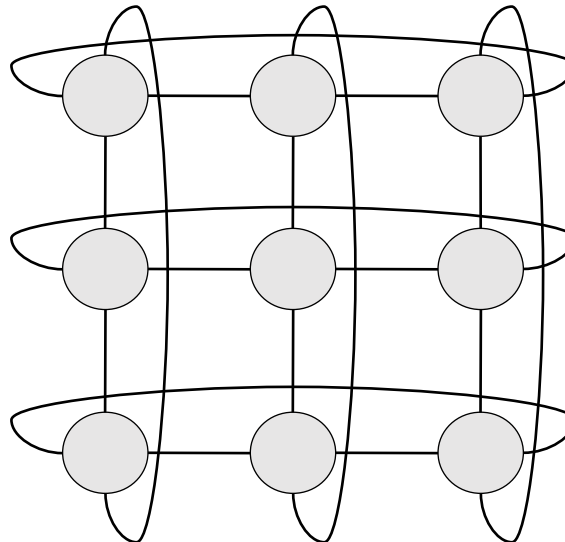
Average Minimal Hop Count:

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Network Diameter:

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- Pair of source and destination nodes that are furthest apart



Network diameter? 2

Avg min hop count? 1.33

Network-On-Chip – Maximum Channel Load

Maximum Channel Load:

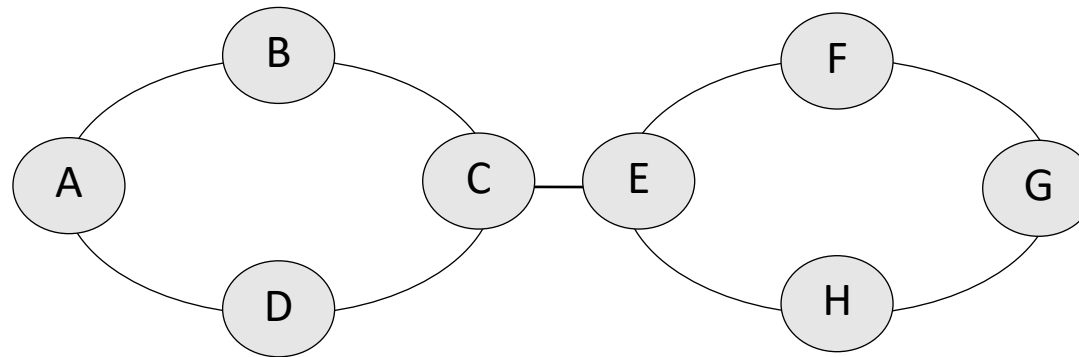
Amount of traffic (bits/cycle) going through the most congested (unidirectional) link in the network.

- Proxy for estimating saturation throughput (i.e., maximum traffic that can be injected)
- Affected by topology, routing and flow control
- Assume all nodes inject max traffic, find most congested link and estimate its load

Network-On-Chip – Maximum Channel Load

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
- Uniform random traffic: all nodes have equal probability of sending to all nodes

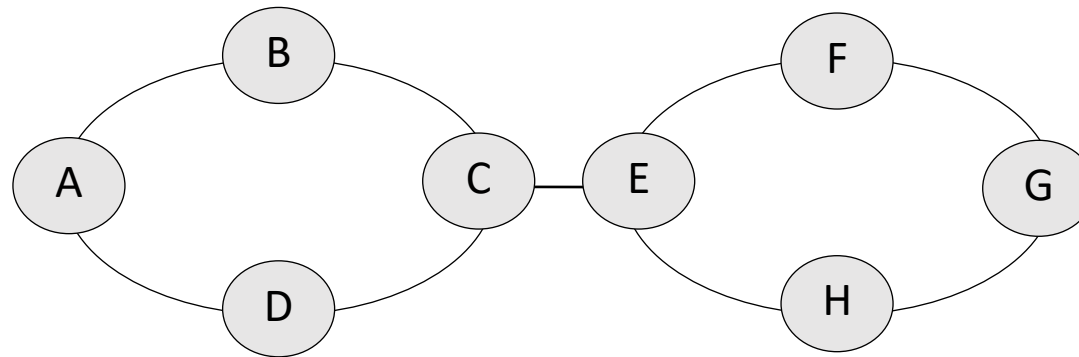


***What is the maximum channel load?
(And at what fraction of max injection bandwidth will the network saturate?)***

Network-On-Chip – Maximum Channel Load

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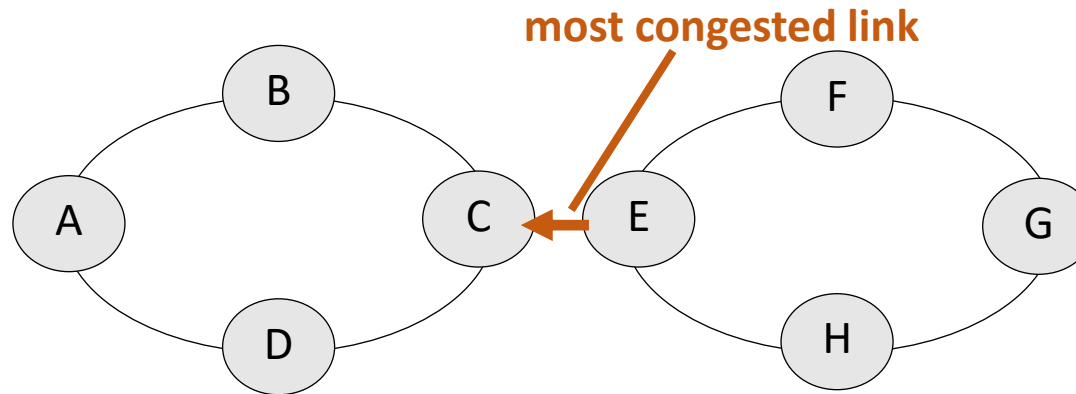


Each cycle, there is $1/8$ probability that the packet injected by node X is destined for node Y.

Network-On-Chip – Maximum Channel Load

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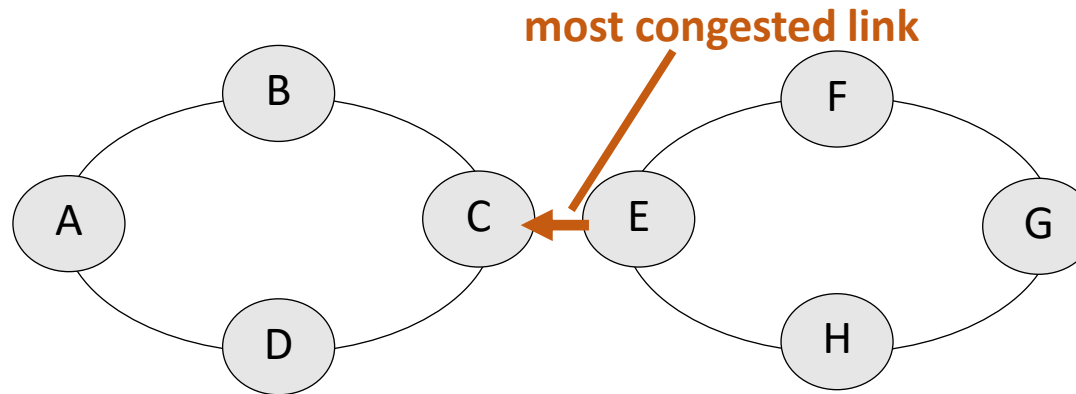


Each cycle, there is $1/8$ probability that the packet injected by node X is destined for node Y.
Each cycle, $1/2$ of each node's injected packets are destined to traverse the most congested link.
Therefore, the maximum channel load is 2 packets per cycle.

Network-On-Chip – Maximum Channel Load

If each (unidirectional) link can only handle max load of 1 packet per cycle:

- Then the network saturates at 1/2 of the max injection bandwidth (i.e., all nodes inject 1 packet per 2 cycles)



Each cycle, there is 1/8 probability that the packet injected by node X is destined for node Y.
Each cycle, 1/2 of each node's injected packets are destined to traverse the most congested link.
Therefore, the maximum channel load is 2 packets per cycle.

Network-On-Chip – Bisection Bandwidth

Network Cut:

Partition that splits all nodes into two disjoint sets

Network Bisection:

Cut with the same number of nodes in each set (or nearly the same)

Bisection Bandwidth:

Minimum cross-cut bandwidth among all bisections

- Proxy for estimating cost (i.e., higher bisection bandwidth implies more global wiring)
- Affected by topology, routing and flow control

Bisection Traffic:

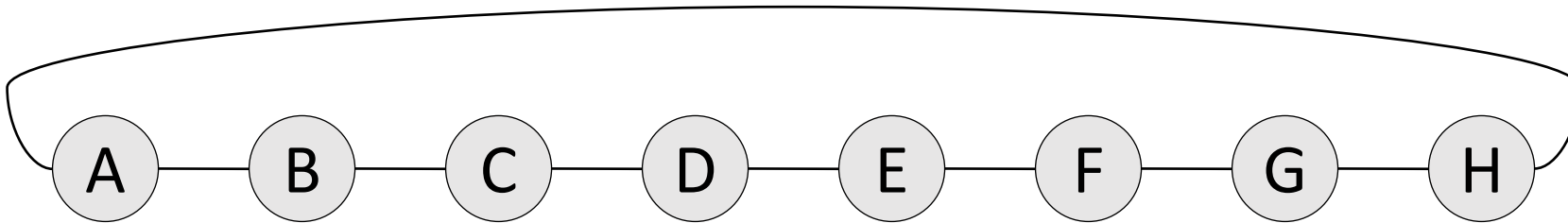
Amount of traffic crossing minimum bisection

- Proxy for estimating saturation throughput

Network-On-Chip – Bisection Bandwidth

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count

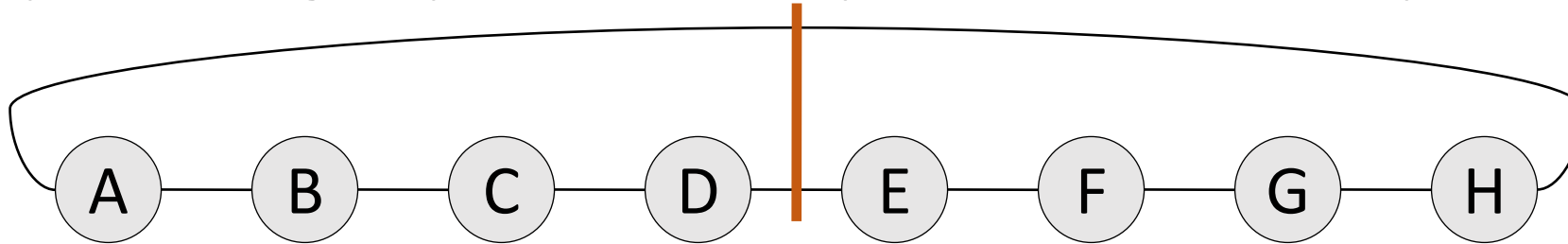


What is the bisection bandwidth?

Network-On-Chip – Bisection Bandwidth

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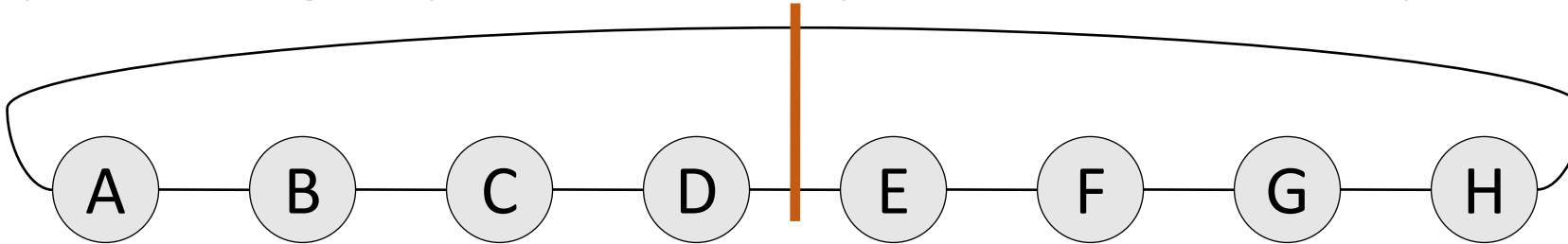
What is the bisection bandwidth?

4 unidirectional links

Network-On-Chip – Bisection Traffic

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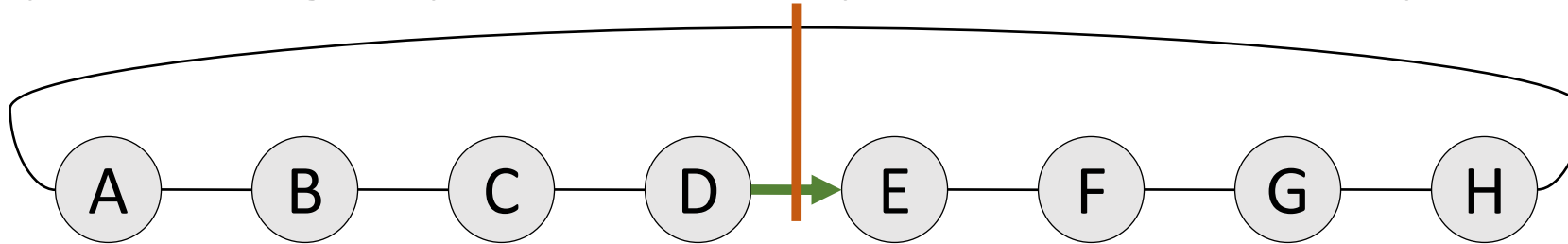


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Network-On-Chip – Bisection Traffic

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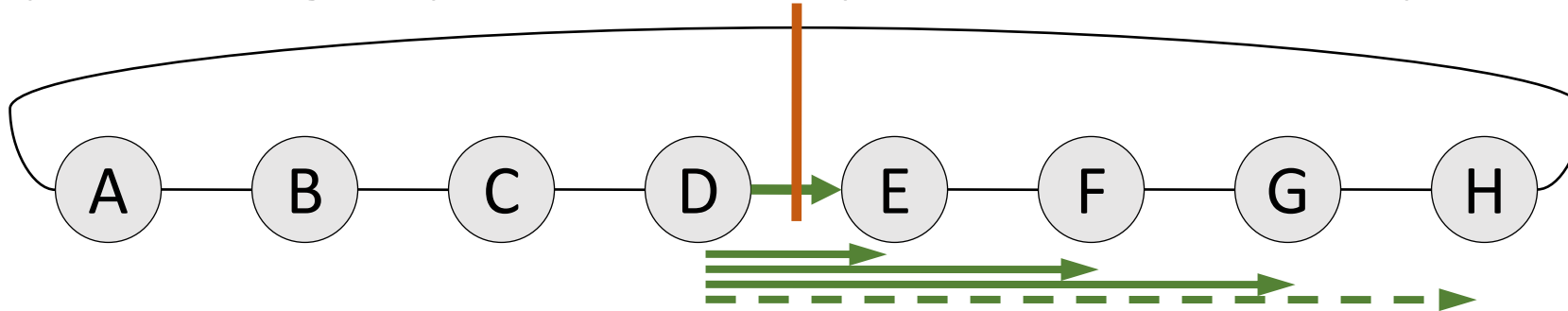


At link from D to E:

Network-On-Chip – Bisection Traffic

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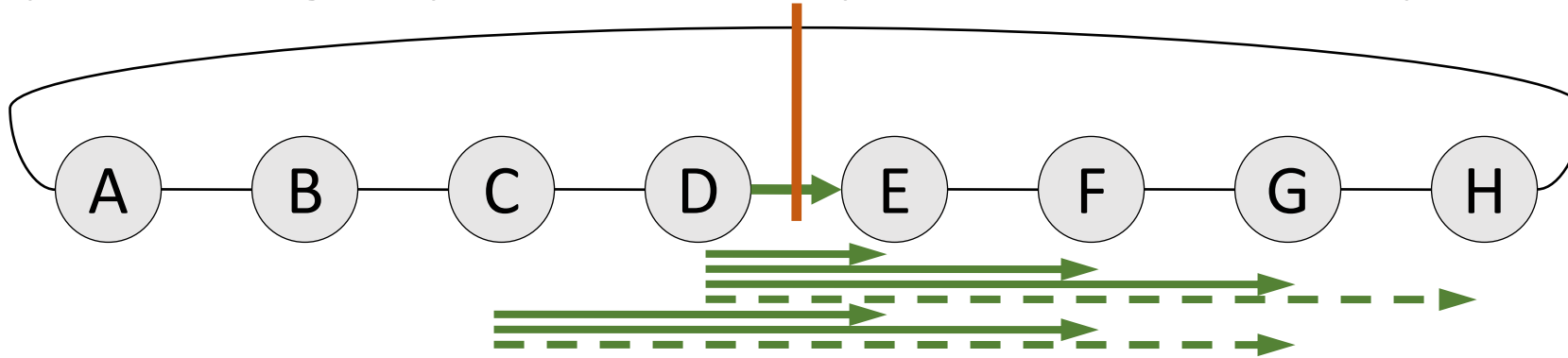
At link from D to E:

- D sends $1/8$ of its traffic to E, F, G and $1/16$ of its traffic to H (since 2 possible shortest paths to H).

Network-On-Chip – Bisection Traffic

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- Max injection bandwidth: all nodes inject 1 packet per cycle
- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count



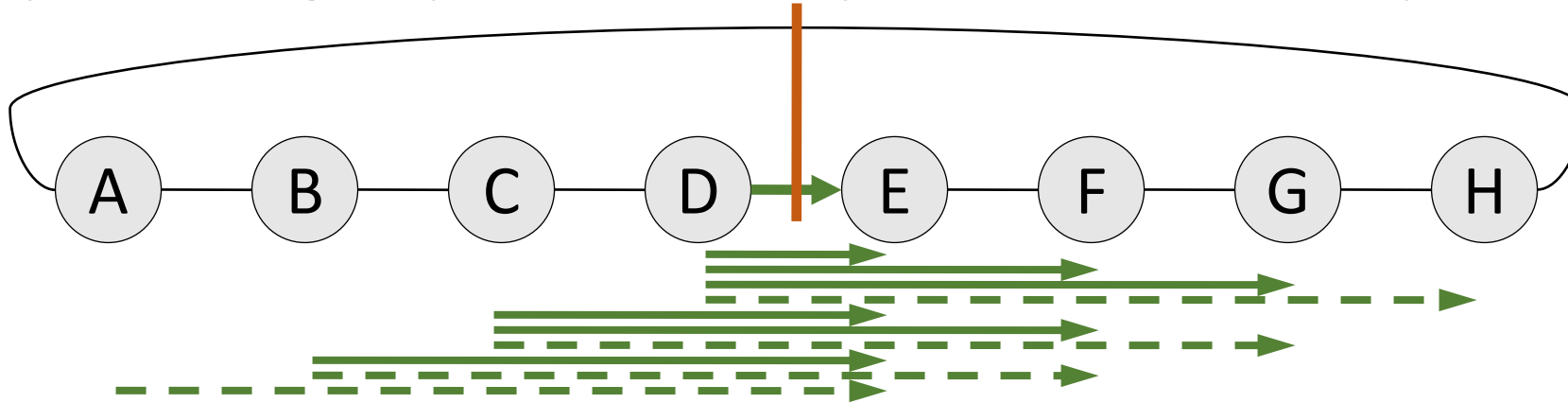
At link from D to E:

- D sends $1/8$ of its traffic to E, F, G and $1/16$ of its traffic to H (since 2 possible shortest paths to H).
- C sends $1/8$ of its traffic to E, F and $1/16$ of its traffic to G.

Network-On-Chip – Bisection Traffic

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
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At link from D to E:

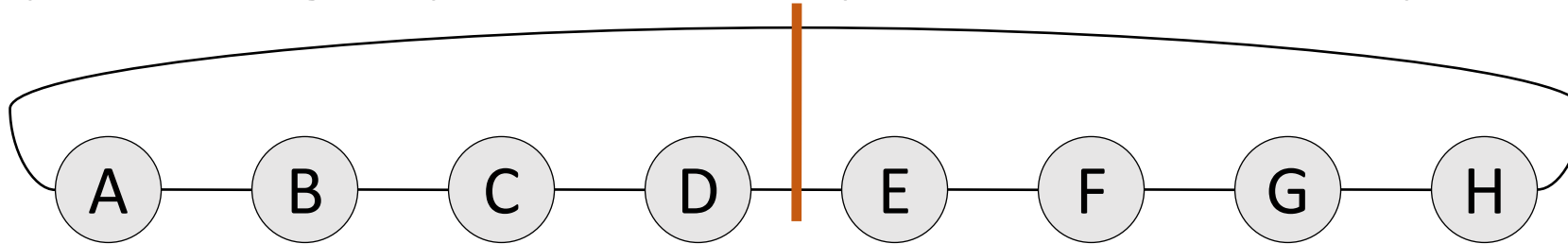
- D sends $1/8$ of its traffic to E, F, G and $1/16$ of its traffic to H (since 2 possible shortest paths to H).
- C sends $1/8$ of its traffic to E, F and $1/16$ of its traffic to G.
- B sends $1/8$ of its traffic to E and $1/16$ to F, and A sends $1/16$ to E.

Therefore, channel load of link from D to E is 1 packet per cycle.

Network-On-Chip – Bisection Traffic

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count



Channel load of link from D to E is 1 packet per cycle.

Channel load of link from E to D is 1 packet per cycle.

Channel load of link from A to H is 1 packet per cycle.

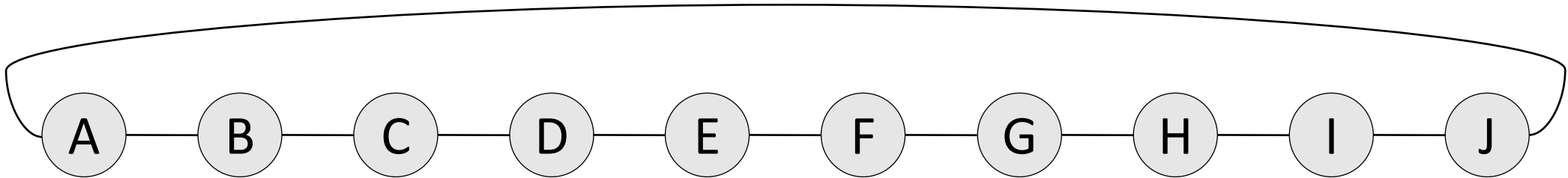
Channel load of link from H to A is 1 packet per cycle.

Therefore, bisection traffic is 4 packets per cycle.

Network-On-Chip – Bisection Bandwidth

e.g., Assume:

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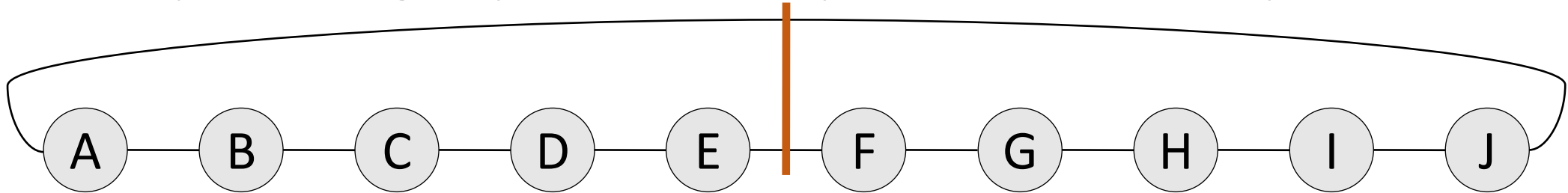


What is the bisection bandwidth?

Network-On-Chip – Bisection Bandwidth

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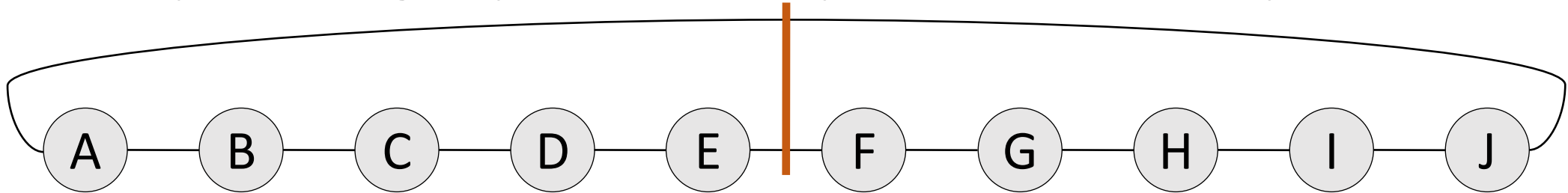
What is the bisection bandwidth?

4 unidirectional links

Network-On-Chip – Bisection Traffic

e.g., Assume:

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- Shortest-path routing: all packets traverse path with minimum hop count

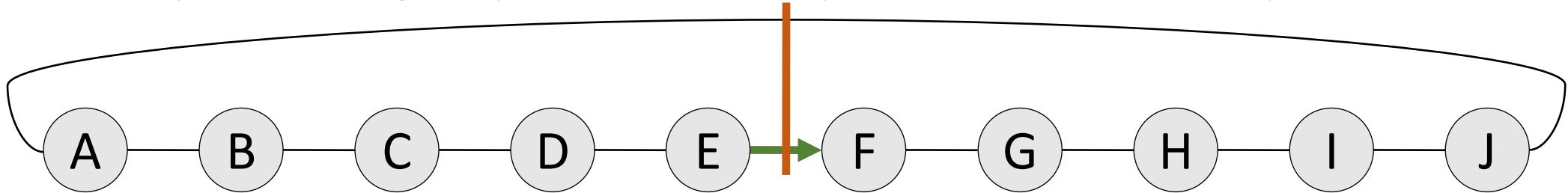


What is the bisection traffic?

Network-On-Chip – Bisection Traffic

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- Max injection bandwidth: all nodes inject 1 packet per cycle
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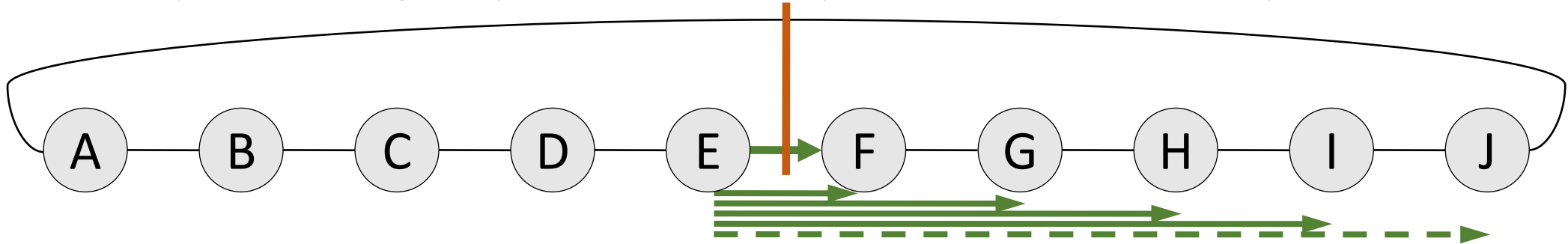


At link from E to F:

Network-On-Chip – Bisection Traffic

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
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- Shortest-path routing: all packets traverse path with minimum hop count



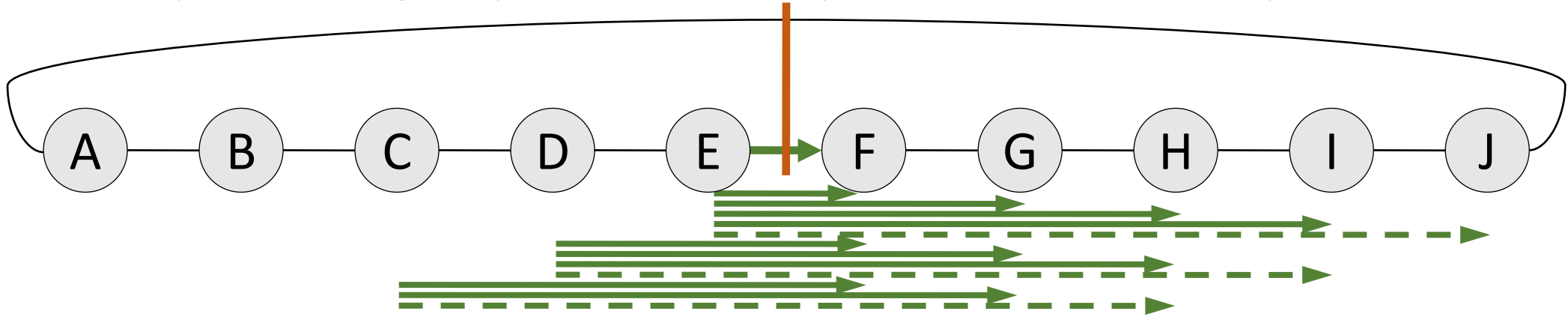
At link from E to F:

- E sends $1/10$ of its traffic to F, G, H, I and $1/20$ to J.

Network-On-Chip – Bisection Traffic

e.g., Assume:

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- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count



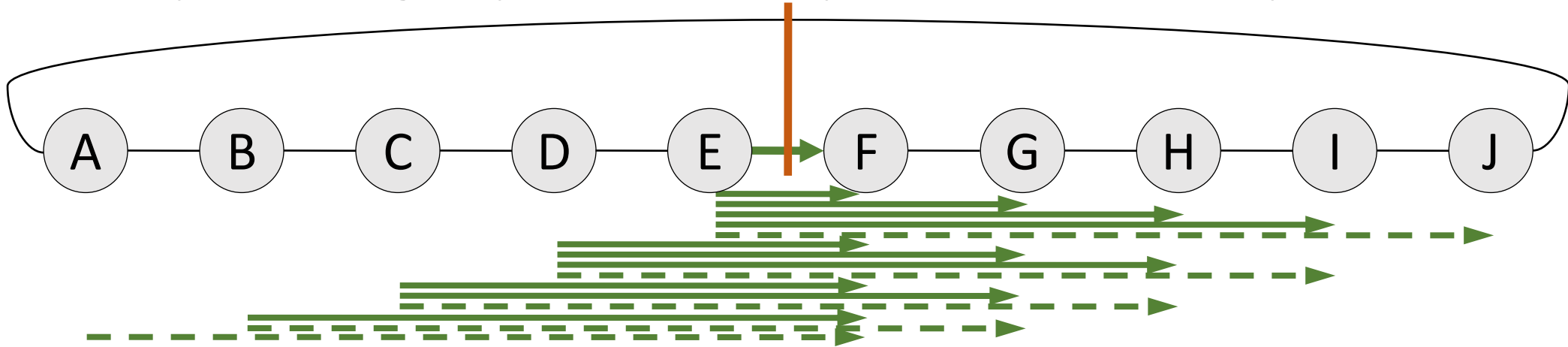
At link from E to F:

- E sends $1/10$ of its traffic to F, G, H, I and $1/20$ to J.
- D sends $1/10$ of its traffic to F, G, H and $1/20$ to I. C sends $1/10$ to F, G and $1/20$ to H.

Network-On-Chip – Bisection Traffic

e.g., Assume:

- Max injection bandwidth: all nodes inject 1 packet per cycle
- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count



At link from E to F:

- E sends $1/10$ of its traffic to F, G, H, I and $1/20$ to J.
- D sends $1/10$ of its traffic to F, G, H and $1/20$ to I. C sends $1/10$ to F, G and $1/20$ to H.
- B sends $1/10$ to F and $1/20$ to G. A sends $1/20$ to F.

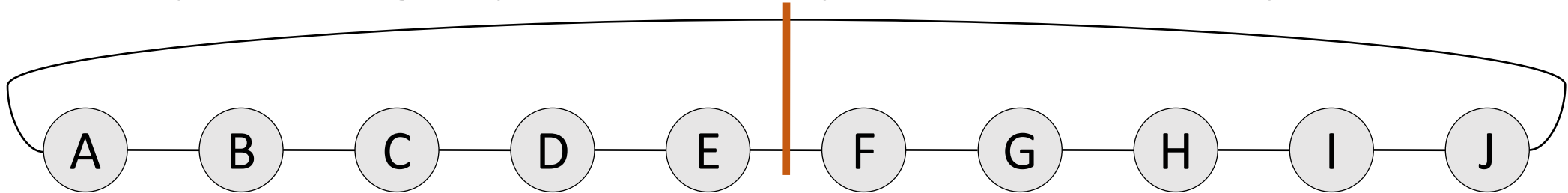
Therefore, channel load of link from E to F is 1.25 packets per cycle.



Network-On-Chip – Bisection Traffic

e.g., Assume:

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- Uniform random traffic: all nodes have equal probability of sending to all nodes
- Shortest-path routing: all packets traverse path with minimum hop count

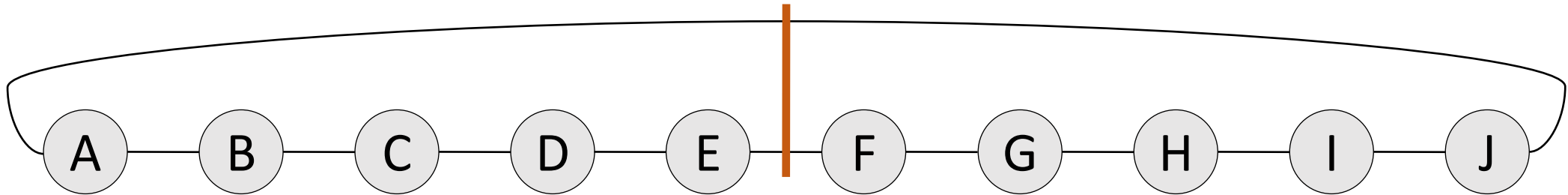


Channel load of link from E to F is 1.25 packets per cycle.
Channel load of link from F to E is 1.25 packets per cycle.
Channel load of link from A to J is 1.25 packets per cycle.
Channel load of link from J to A is 1.25 packets per cycle.
Therefore, bisection traffic is 5 packets per cycle.

Network-On-Chip – Bisection Traffic

Since bisection bandwidth is only 4 packets per cycle:

- Then the network saturates at $4/5$ of the max injection bandwidth (i.e., all nodes inject 4 packets per 5 cycles)



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Channel load of link from A to J is 1.25 packets per cycle.
Channel load of link from J to A is 1.25 packets per cycle.
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Network-On-Chip – Path Diversity

Path Diversity:

Number of possible paths between each source-destination pair

- Proxy for estimating fault tolerance and load balancing
- Affected by topology and routing

Network-On-Chip – Path Diversity

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Network-On-Chip – Path Diversity

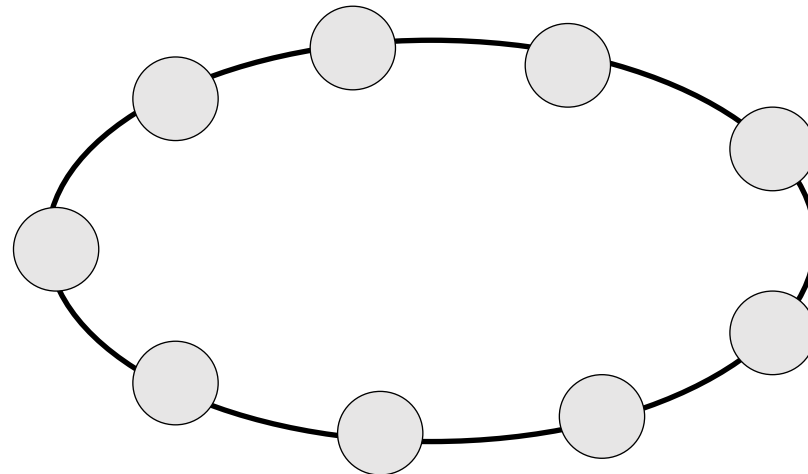
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Path diversity?

Network-On-Chip – Path Diversity

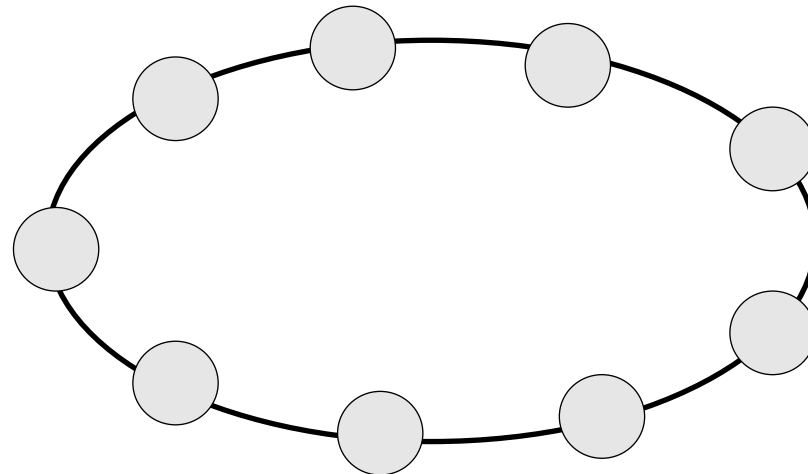
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Path diversity? 1

Network-On-Chip – Path Diversity

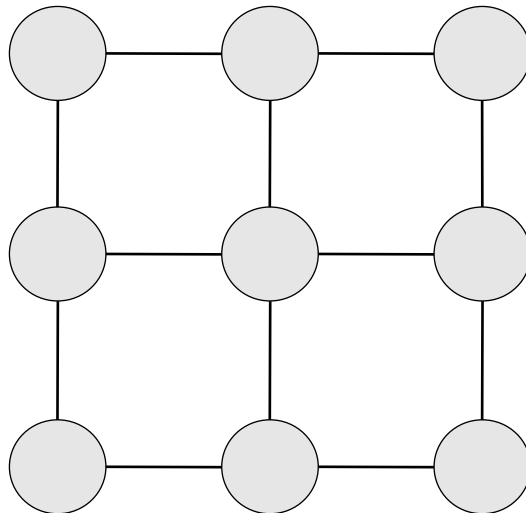
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Min path diversity?
Max path diversity?

Network-On-Chip – Path Diversity

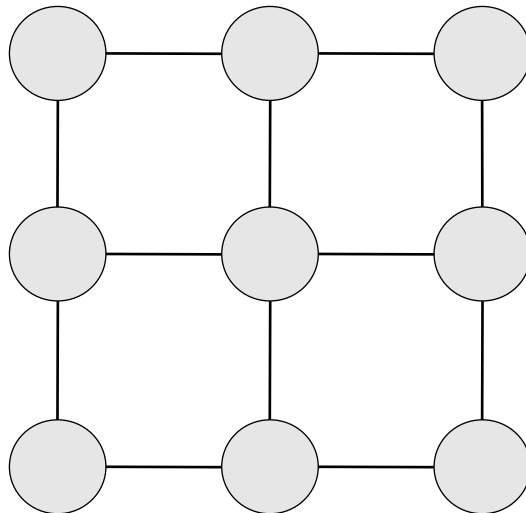
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Min path diversity? 1

Max path diversity?

Network-On-Chip – Path Diversity

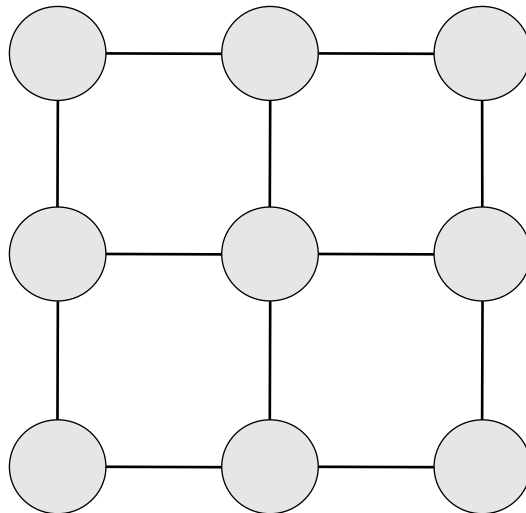
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Min path diversity? 1

Max path diversity? 6

Network-On-Chip – Path Diversity

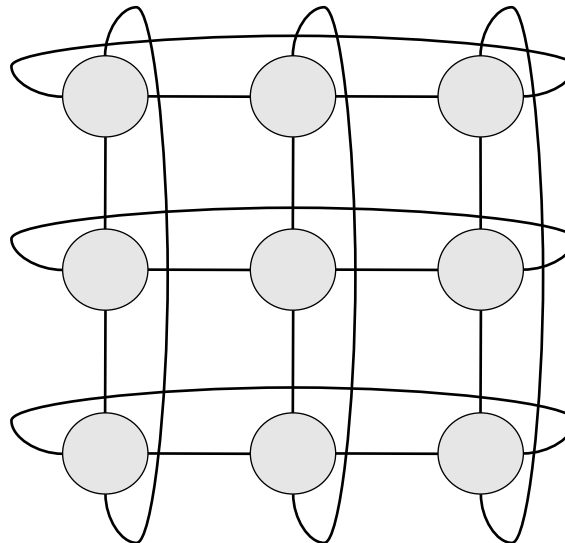
Path Diversity:

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- Affected by topology and routing

e.g., Assume:

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Max path diversity?

Network-On-Chip – Path Diversity

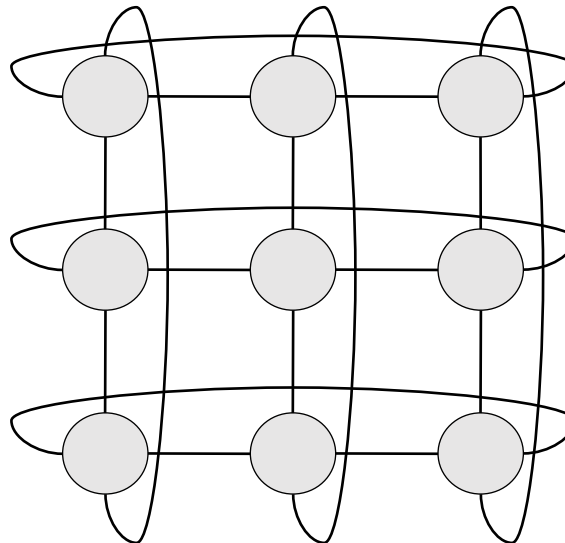
Path Diversity:

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- Affected by topology and routing

e.g., Assume:

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Max path diversity? 2

Network-On-Chip – Path Diversity

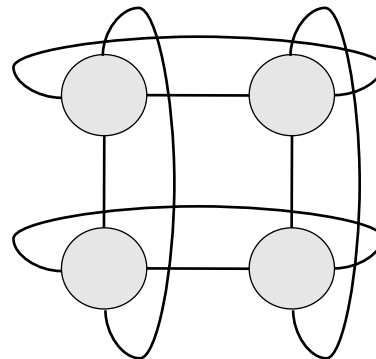
Path Diversity:

Number of possible paths between each source-destination pair

- Proxy for estimating fault tolerance and load balancing
- Affected by topology and routing

e.g., Assume:

- Shortest-path routing: all packets traverse path with minimum hop count



Max path diversity?

Network-On-Chip – Path Diversity

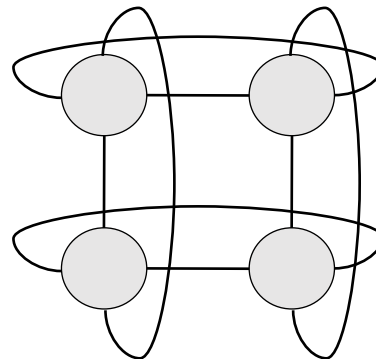
Path Diversity:

Number of possible paths between each source-destination pair

- Proxy for estimating fault tolerance and load balancing
- Affected by topology and routing

e.g., Assume:

- Shortest-path routing: all packets traverse path with minimum hop count



Max path diversity? 8